

POLD1 fusion-hotspot benchmark report

Study: msk_impact_50k_2026

Abstract

This report analyzes POLD1 gene fusions from the msk_impact_50k_2026 study, covering 16 fusion events. 2/16 fusions (12.5%) were in-frame. 6/16 fusions (37.5%) retained the PF00136 domain. Domain retention was tested with Fisher's exact test ($p=0.192308$, not statistically significant at $\alpha=0.05$). No literature benchmark is configured for POLD1.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 16 analyzed fusion events, identifying 15 distinct fusion partner genes. The most frequent partner was MYH14, observed in 2 events.

Domain retention

PF00136 domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 2/2 (100.0%) of in-frame fusions retained the domain, $p=0.192308$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.435644.

Domain disruption

Disruption of the configured disruption-required domain(s) was tested with Fisher's exact test comparing in-frame fusions against all others; 1/2 (50.0%) of in-frame fusions disrupted the domain, $p=0.871795$ (not statistically significant at $\alpha=0.05$). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.910891.

Cutpoint detection

Cutpoint detection scanned 13 mapped breakpoints for the protein position that best separates domain-retained from lost/disrupted fusions; the inferred cutpoint was position 415 aa (permutation-corrected $p=0.00990099$, statistically significant at $\alpha=0.05$). This is -62 aa from the nearest configured domain boundary at 477 aa.

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" ($n=6$) and "not_retained" ($n=7$). For Frame_status, retained: 2/3 (66.7%, 95% CI 20.8%-93.9%); not_retained: 0/3 (0.0%, 95% CI 0.0%-56.1%). Welch's t-test comparing Tumor_variant_count between groups gave means of 24.8333 vs 43, $p=0.299234$ (not statistically significant at $\alpha=0.05$).

Composite evidence score

Composite evidence score produced results for this run (15 row(s) in composite_evidence_ranking); see the accompanying table.

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Joint partner

Joint partner reported: POLD1 has no gene_pair configured; joint-partner analysis was skipped.

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
MYH14	2	0.125	0.03608686972964552	0.004053354643708729	1.0	0.3538272568836993	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.23271511404952305	1
TMEM87A	1	0.0625	0.03608686972964552	0.004053354643708729	0.8578947368421053	0.3538272568836993	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.19264932457583883	2
NR1H2	1	0.0625	0.03608686972964552	0.004053354643708729	0.7157894736842105	0.3538272568836993	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.17133353510215465	3
ZNF284	1	0.0625	0.03608686972964552	0.004053354643708729	0.6210526315789473	0.3538272568836993	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.15712300878636515	4
LILRB1	1	0.0625	0.03608686972964552	0.004053354643708729	0.5859649122807018	0.3538272568836993	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.15185985089162832	5
NDUFA3	1	0.0625	0.03608686972964552	0.004053354643708729	0.5701754385964912	0.3538272568836993	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.14949142983899674	6
PNPLA6	1	0.0625	0.03608686972964552	0.004053354643708729	0.4666666666666667	0.3538272568836993	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.13396511404952305	7
PEPD	1	0.0625	0.03608686972964552	0.004053354643708729	0.3298245614035088	0.3538272568836993	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11343879826004938	8
SIGLECL1	1	0.0625	0.03608686972964552	0.004053354643708729	0.3070175438596491	0.3538272568836993	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.11001774562847041	9

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
C5AR1	1	0.0625	0.03608686972964552	0.004053354643708729	0.3035087719298246	0.3538272568836993	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.10949142983899675	10
SRRM1	1	0.0625	0.03608686972964552	0.004053354643708729	0.07719298245614037	0.3538272568836993	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.07554406141794412	11
ADAMTSL1	1	0.0625	0.03608686972964552	0.004053354643708729	None	0.3538272568836993	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.07525307535238007	12
CIC	1	0.0625	0.03608686972964552	0.004053354643708729	None	0.3538272568836993	['confidence_certainty', 'domain_disruption', 'domain_retention', 'recurrence']	0.07525307535238007	13
KLK10	1	0.0625	0.03608686972964552	0.004053354643708729	0.01754385964912286	0.3538272568836993	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.06659669299689149	14
SPIB	1	0.0625	0.03608686972964552	0.004053354643708729	0.0	0.3538272568836993	['confidence_certainty', 'cutpoint_proximity', 'domain_disruption', 'domain_retention', 'recurrence']	0.06396511404952306	15

cutpoint_detection tables

cutpoint_scan

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
18	1	5	0	7	None	0.46153846153846156	0.3357921019231931
170	2	4	0	7	None	0.1923076923076923	0.7160033436347992
179	3	3	0	7	None	0.06993006993006992	1.1553360374650619
253	4	2	0	7	None	0.02097902097902098	1.6782147827453993
415	5	1	0	7	None	0.004662004662004663	2.3314272965207428
496	5	1	1	6	30.0	0.029137529137529136	1.5355472791766678
631	5	1	2	5	12.5	0.10256410256410255	0.9890046156985369
719	5	1	3	4	6.666666666666667	0.26573426573426573	0.5755524408482516
797	5	1	4	3	3.75	0.5594405594405595	0.2522460504731182
810	5	1	5	2	2.0	1.0	-0.0
941	5	1	6	1	0.8333333333333334	1.0	-0.0

cutpoint	n_positive_at_or_below	n_positive_above	n_negative_at_or_below	n_negative_above	odds_ratio	p_value	neg_log10_p_value
975	5	1	7	0	0.0	0.46153846153846156	0.3357921019231931

domain_disruption tables

frame_domain_contingency_table

1	7
1	4

domain_disruption_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_disruption_pf03104	DNA polymerase family B, exonuclease domain	13	5	4	4	0.3224063400576369

domain_retention tables

frame_domain_contingency_table

2	4
0	7

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
auto_key_domain	DNA polymerase family B	13	6	6	1	0.2753912753912754

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
13	8	16	0.5

frequency tables

Partner_gene_counts

Partner_gene	Event_count
ADAMTSL1	1

Partner_gene	Event_count
C5AR1	1
CIC	1
KLK10	1
LILRB1	1
MYH14	2
NDUFA3	1
NR1H2	1
PEPD	1
PNPLA6	1
SIGLECL1	1
SPIB	1
SRRM1	1
TMEM87A	1
ZNF284	1

Per-event results (results.tsv)

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-0069297-T01-IM7-1	P-0069297-T01-IM7		C5AR1-POLD1	C5AR1	unknown	three_prime	50902161	2	18	False	retained	1.0	False	DNA polymerase family B; DNA polymerase family B, exonuclease domain			C5AR1 (NM_001736) - POLD1 (NM_002691) rearrangement: c.3+879:C5AR1_c.53:POLD1del Protein Fusion: mid-exon {C5AR1:POLD1}	NA
EVT-P-0019604-T01-IM6-3	P-0019604-T01-IM6		MYH14-POLD1	MYH14	in-frame	three_prime	50909319	11	415	True	retained	1.0	False	DNA polymerase family B		DNA polymerase family B, exonuclease domain	MYH14 (NM_001145809) - POLD1 (NM_002691) Rearrangement : c.4539+61:MYH14_c.1243-120:POLD1del Protein Fusion: in frame {MYH14:POLD1}	NA

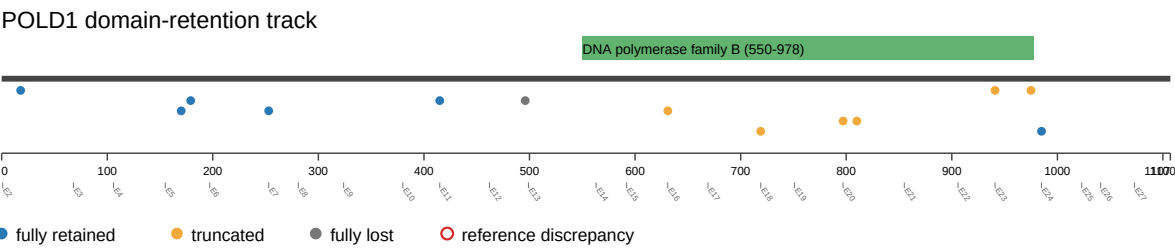
event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-005 1640-T01-I M6-5	P-0051 640-T0 1-IM6		NR1H2-POLD1	NR1H2	out-of-frame	three_prime	50905674	7	253	True	retained	1.0	False	DNA polymerase family B		DNA polymerase family B, exonuclease domain	NR1H2 (NM_007121) - POLD1 (NM_002691) rearrangement: c.1028-692:NR1H2_c.759-37:POLD1del Protein Fusion: out of frame {NR1H2:POLD1}	NA
EVT-P-005 2327-T01-I M6-6	P-0052 327-T0 1-IM6		PEPD-POLD1	PEPD	out-of-frame	three_prime	50917974	20	797	True	disrupted	0.424242424242425	True		DNA polymerase family B, exonuclease domain	DNA polymerase family B	PEPD (NM_000285) - POLD1 (NM_002691) rearrangement: c.967+204:PEPD_c.2389-98:POLD1inv Protein Fusion: out of frame {PEPD:POLD1}	NA
EVT-P-000 5493-T01-I M5-7	P-0005 493-T0 1-IM5		PNPLA6-POLD1	PNPLA6	unknown	three_prime	50916627	18	719	True	disrupted	0.6060606060606061	True		DNA polymerase family B, exonuclease domain	DNA polymerase family B	PNPLA6 (NM_001166111) - POLD1 (NM_002691) rearrangement: c.1323:PNPLA6_c.2155-56:POLD1del Protein fusion: mid-exon (PNPLA6-POLD1)	NA
EVT-P-003 2778-T01-I M6-9	P-0032 778-T0 1-IM6		POLD1-KLK10	KLK10	unknown	three_prime	50919757	23	975	False	disrupted	0.009324009324009324	True		DNA polymerase family B, exonuclease domain	DNA polymerase family B	KLK10 (NM_001077500) - POLD1 (NM_002691) Rearrangement : c.88+823:KLK10_c.2925:POLD1inv Protein Fusion: mid-exon {KLK10:POLD1}	NA
EVT-P-000 3038-T01-I M5-10	P-0003 038-T0 1-IM5		POLD1-LILRB1	LILRB1	unknown	three_prime	50905327	5	179	False	retained	1.0	False	DNA polymerase family B		DNA polymerase family B, exonuclease domain	NA Protein fusion: mid-exon (LILRB1-POLD1)	NA

event_id	sample_id	patient_id	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	domain_status	domain_retained_fraction	domain_is_truncated	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-P-001 6377-T01-I M6-11	P-0016 377-T0 1-IM6		POLD 1-NDU FA3	ND UFA 3	unknown	three_prime	509053 02	5	170	False	retained	1.0	False	DNA polymerase family B		DNA polymerase family B, exonuclease domain	NDUFA3 (NM_004542) - POLD1 (NM_002691) rearrangement: c.85+584:NDUFA3_c.510:POLD1dup Protein Fusion: mid-exon {NDUFA3:POLD1}	NA
EVT-P-002 3588-T01-I M6-32	P-0023 588-T0 1-IM6		POLD 1-SIG LECL1	SIG LEC L1	unknown	five_prime	509181 12	20	810	False	disrupted	0.6083916 08391608 4	True	DNA polymerase family B, exonuclease domain		DNA polymerase family B	POLD1 (NM_002691) - SIGLECL1(NM_173635) Rearrangement : c.2429:POLD1_c.-191+1669:SIGLECL1del Protein Fusion: mid-exon {POLD1:SIGLECL1}	NA
EVT-P-004 7491-T01-I M6-33	P-0047 491-T0 1-IM6		POLD 1-SPI B	SPI B	in-frame	five_prime	509198 61	23	985	True	retained	1.0	False	DNA polymerase family B; DNA polymerase family B, exonuclease domain			POLD1 (NM_002691) - SPIB (NM_003121) rearrangement: c.2954-6:POLD1_c.490+532:SPIBdel Protein Fusion: in frame {POLD1:SPIB}	NA
EVT-P-001 4626-T01-I M6-34	P-0014 626-T0 1-IM6		POLD 1-SRR M1	SRR M1	out-of-frame	three_prime	509196 23	23	941	True	disrupted	0.0885780 88578088 58	True		DNA polymerase family B, exonuclease domain	DNA polymerase family B	SRRM1 (NM_005839) - POLD1 (NM_002691) rearrangement: t(1;19)(p36.11;q13.33)(chr1:g.24979609::chr19:g.50919623) Protein fusion: out of frame (SRRM1-POLD1)	NA
EVT-P-006 4872-T01-I M7-35	P-0064 872-T0 1-IM7		POLD 1-TME M87A	TME M87 A	unknown	five_prime	509097 66	12	496	False	lost	0.0	False	DNA polymerase family B, exonuclease domain	DNA polymerase family B		POLD1 (NM_002691) - TMEM87A (NM_015497) rearrangement: t(15;19)(q15.1;q13.33)(chr15:g.42511439::chr19:g.50909766) Protein Fusion: mid-exon {POLD1:TMEM87A}	NA

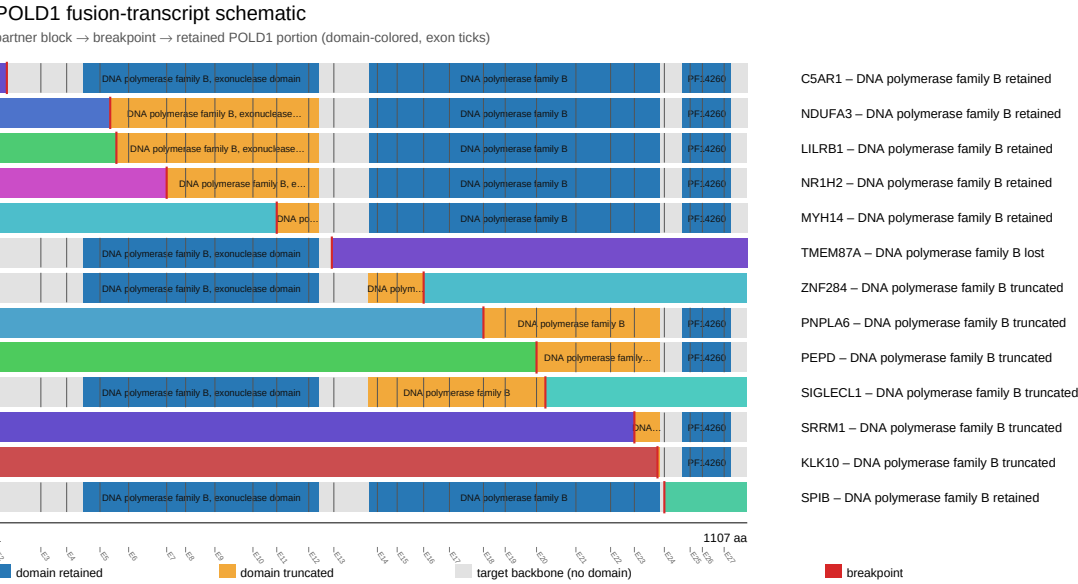
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EVT-P-001 7653-T01-I M5-36	P-0017 653-T0 1-IM5		ZNF28 4-POL D1	ZNF 284	out- of-fr ame	five _pr im e	509122 01	15	631	True	disru pted	0.1911421 91142191 14	True	DNA p olymer ase family B, exo nuclea se do main		DNA p olymer ase family B	POLD1 (NM_002691) - ZNF284 (NM_001037813) rearrangement: c.1892+43: POLD1_c.15+1157: ZNF284dup Protein Fusion: out of frame {POLD1:ZNF284}	NA

Figures

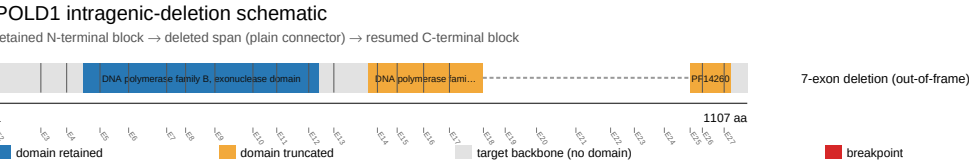
domain retention outliers



fusion schematic



intragenic deletion schematic



reference comparison

Reference vs msk_impact_50k_2026

